

Supply-chain inventory consistency is a degree-one class and is independent of the Zappa–Szép weld obstruction

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Abstract

A supply chain, abstracted to a small category, composes with a second chain sharing a warehouse as a Zappa–Szép product $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$, whose existence is obstructed by a degree-two class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{S}}; \mathcal{D})$. It is tempting — and, for blockchain re-entrancy, correct — to identify “inventory is inconsistent between two routes” with “ $[\omega] \neq 0$ ”. We show this identification fails for supply chains, in both directions. Writing the inventory as a resource presheaf R on $\mathcal{S}$, we prove that the matched-pair law of the Zappa–Szép product is invisible to global sections of R (Lemma 2), so inventory consistency is cut out entirely by the two factors and is the vanishing of a *degree-one* class $[\theta_R] \in H^1(\mathcal{S}; R)$ — the route-reconvergence discrepancy (Theorem 5). This class is logically independent of $[\omega]$: they live in different degrees over different coefficient systems, and we exhibit explicit chains realizing each of $([\omega] = 0, [\theta_R] \neq 0)$ and $([\omega] \neq 0, [\theta_R] = 0)$ (Theorem 8). The conjecture was a false generalization of re-entrancy, where the inventory *is* the weld-state and no separate resource presheaf exists. The payoff is a classification of compositional failure by cohomological degree: the degree records where the conserved quantity lives — on the nodes (H^1 , route discrepancy) or in the weld (H^2 , handoff order).

1 Introduction

Composition is where correctness is won or lost. When two procurement flows merge at a shared warehouse, when two smart contracts interleave at a shared ledger, or when two delivery routes reconverge on the same stock-keeping unit, the question is always the same: do the local data glue to a single consistent global state? When they do not, the failure has an economic price — a double-spend, a phantom inventory, a reconciliation that never closes. A compositional theory earns its keep when it tells you *which* failures can occur and *where* they live.

The categorical model is by now standard. A supply chain, with nodes as objects and admissible logistics operations as morphisms, is a small category; equivalently, by Ahman and Uustalu [1], a *directed container*. Composing two chains that share a set of warehouses — procurement $\mathcal{C}$ followed by logistics $\mathcal{D}$ — amounts to forming a *Zappa–Szép product* $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$, a strict factorization in which every operation of $\mathcal{S}$ decomposes uniquely as a procurement step followed by a logistics step [2, 4]. Such a factorization need not exist; its existence is governed by a cohomological obstruction $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{S}}; \mathcal{D})$ in the sense of Baues–Wirsching [3], which vanishes exactly when the composite

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chain closes into the claimed product [7]. In blockchain re-entrancy the same degree-two class, specialized to $\mathbb{Z}/2$, is the double-spend bit: $[\omega] = \varepsilon$, a fact we have verified in Lean [8].

This invites a conjecture, and it is the conjecture we test.

(C) Global inventory is consistent — the resource data glue to a route-independent stock over the whole chain — if and only if the Zappa–Szép weld obstruction $[\omega] \in H^2(\text{Sk}_S; \mathcal{D})$ vanishes.

It is a reasonable conjecture. It reads inventory inconsistency as a failure of the chain to compose, and it is exactly what re-entrancy would lead you to expect. But it conflates two things that a careful bookkeeping keeps apart: *whether the chain factors at all* and *whether, given the operations, stock reconciles across routes*. No one had written down the resource data as an explicit presheaf and compared its gluing obstruction to $[\omega]$. When one does, the conjecture collapses.

Results. We model inventory as a resource presheaf $R: \mathcal{S}^{\text{op}} \rightarrow \mathbf{Ab}$ — a local system of stock-groups transported along operations — and define inventory consistency as the existence of a global section. Three results follow.

- **The weld is invisible to inventory** (Lemma 2). A family of local stock values is a global section of R over $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ if and only if it is compatible with every $\mathcal{C}$ -operation and every $\mathcal{D}$ -operation; the matched-pair law λ that defines the Zappa–Szép product plays no role. Inventory consistency is the degree-one amalgamation problem for the cover $\{\mathcal{C}, \mathcal{D}\}$ and nothing more.
- **Inventory consistency is a degree-one class** (Theorem 5). Global inventory is consistent if and only if a degree-one class $[\theta_R] \in H^1(\mathcal{S}; R)$ vanishes. When the two factors are acyclic, Mayer–Vietoris identifies this class with the *reconvergence discrepancy*: the failure of the node-potentials determined on each factor to agree on the shared warehouses.
- **The two classes are independent** (Theorem 8). Both $([\omega] = 0, [\theta_R] \neq 0)$ and $([\omega] \neq 0, [\theta_R] = 0)$ occur. Hence neither implication of (C) holds. The classes live in different degrees over different coefficient systems (R versus $\mathcal{D}$), so there is no natural map identifying them.

Why it looked true, and why there is no rescue. Re-entrancy is the degenerate case in which the conserved quantity *is* the weld-state — the composite protocol’s own state, living inside the matched pair — so there is no separate resource presheaf and $[\omega]$ is the whole story. Conjecture (C) silently imports this identification into supply chains, where stock lives on the *nodes* and is a genuinely separate degree-one datum (Proposition 10). This also reconciles an earlier worked example (§6): the family $W_{n,\varepsilon}$ that computes $[\omega] = \varepsilon$ measures a loop built from *parallel* arrows, which closes inside a single hom-set and is genuinely a weld phenomenon — a correct computation of the *other* class.

The payoff. The honest result is more useful than (C) would have been. It classifies compositional failure by cohomological degree, keyed to the locus of the conserved quantity (§7): stock on the nodes gives an H^1 route discrepancy (supply-chain inventory), a conserved quantity in the weld gives an H^2 handoff obstruction (re-entrancy), and the factorization itself existing is a third H^2 condition. Two distinct economic failure modes, cleanly separated by degree.

Outline. Section 2 recalls directed containers, the Zappa–Szép weld obstruction, and the resource presheaf, and fixes the one modelling choice the argument depends on. Section 3 proves that the matched-pair law is invisible to inventory. Section 4 establishes the degree-one characterization and its closed form. Section 5 proves independence. Section 6 explains the provenance of (C) and reconciles the earlier model. Section 7 states the degree classification and the open boundary.

Scope. We fix an abstraction — supply chain as free category on an operation quiver, modulo imposed welds — and analyze it honestly; we do not claim it is the unique faithful model of any particular logistics system. Everything below is about that abstraction.

2 Preliminaries

We work with small categories and their cohomology. We recall only what the proofs use.

2.1 Supply chains as directed containers

A *directed container* is a container equipped with the structure making its extension a comonad; by Ahman and Uustalu [1] directed containers are precisely small categories. We use the category presentation throughout: a supply chain is a small category $\mathcal{S}$ whose objects are nodes (warehouses, plants, retailers), whose morphisms out of a node a are the admissible logistics operations starting at a , and whose composition is sequencing of operations. We write $\text{Sk}_{\mathcal{S}}$ for the skeleton (one object per isomorphism class), on which the cohomology of §2.2 is computed.

Modelling choice (essential). We model a supply chain as the *free category on its operation quiver*, modulo only the welds we explicitly impose. In particular two routes $a \rightarrow d$ through distinct intermediate nodes are *distinct morphisms* unless a relation forces them equal. This is the honest object: a chain in which reconvergent routes carry possibly different cost/stock transforms. A thin poset, by contrast, forces $rp = sq$ and makes route discrepancy invisible by fiat; that is not a supply chain but the assertion that all routes are already reconciled. The distinction is load-bearing: witness W1 of Theorem 8 depends on distinct routes being distinct morphisms.

2.2 The Zappa–Szép product and its weld obstruction

A *Zappa–Szép product* (strict factorization) $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ consists of wide subcategories $\mathcal{C}, \mathcal{D} \subseteq \mathcal{S}$ with $\mathcal{C} \cap \mathcal{D} = \mathcal{O}$ the discrete category on the shared object set, such that every morphism of $\mathcal{S}$ factors *uniquely* as $f = c \cdot d$ with $c \in \mathcal{C}, d \in \mathcal{D}$. The product is governed by a matched-pair law, the distributive rewriting $\lambda: \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}, d \cdot c = (c \triangleright d) \cdot (c \triangleleft d)$, in the sense of Rosebrugh–Wood [4]; for directed containers this is the distributive law of Ahman and Uustalu [2].

Such a factorization need not exist. By [6], $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ exists over a chosen wide $\mathcal{D}$ iff a freeness condition (L) and a global-closure condition (G) hold; and by [7], under the abelian regime (H) the defect of (G) is a normalized Baues–Wirsching 2-cocycle ω_T on $\text{Sk}_{\mathcal{C}}$ with coefficients in the vertex-group presheaf $\mathcal{D}: \text{Sk}_{\mathcal{C}}^{\text{op}} \rightarrow \mathbf{Ab}$, so that

$$(G) \text{ holds} \iff [\omega] = 0 \in H^2(\text{Sk}_{\mathcal{S}}; \mathcal{D}).$$

Three features of $[\omega]$ matter below: it depends only on the category $\mathcal{S}$ and the chosen right factor $\mathcal{D}$ (no resource data enter its definition); its coefficient system is $\mathcal{D}$, the vertex groups of the invertible/weld part; and it is a *degree-two* class.

2.3 Inventory as a resource presheaf

Inventory is data carried *on top of* the category: a presheaf $R: \mathcal{S}^{\text{op}} \rightarrow \mathbf{Ab}$, where $R(a)$ is the group of stock configurations at node a and $R(f): R(\text{cod } f) \rightarrow R(\text{dom } f)$ transports stock back along the operation f . This is a section of the container-logic fibration $\text{Cont}(\text{cod}) = \text{Fam}(\text{cod}^{\text{op}})$ [10]: fibrewise, a choice of stock object and its reindexing along operations.

Definition 1 (inventory consistency). A *global section* of R is a family $(x_a \in R(a))_{a \in \text{Ob } \mathcal{S}}$ with $R(f)(x_{\text{cod } f}) = x_{\text{dom } f}$ for every morphism f . Inventory R is *globally consistent* iff it admits a global section — a route-independent assignment of stock to every node, compatible with every operation.

We single out the case that models physical stock transport: a *transport local system* is an R in which each $R(f)$ is translation by an element $\delta_f \in A$ of a fixed abelian group A (so $R(a) = A$ for all a , and δ is additive along composites). Then a global section is a node-potential $\varphi: \text{Ob } \mathcal{S} \rightarrow A$ with $\delta_f = \varphi(\text{cod } f) - \varphi(\text{dom } f)$ for every f ; that is, the 1-cochain δ is a coboundary. Its obstruction is the class

$$[\theta_R] := [\delta] \in H^1(\mathcal{S}; R).$$

Contrast with $[\omega]$: $[\theta_R]$ depends on the resource presheaf R , not on how $\mathcal{S}$ factors; its coefficient system is R (stock groups), not $\mathcal{D}$; and it is a *degree-one* class.

3 The matched-pair law is invisible to inventory

The first observation is that the entire Zappa–Szép structure — the law λ that makes $\mathcal{C} \bowtie \mathcal{D}$ more than a disjoint union — contributes nothing to the inventory gluing problem.

Lemma 2 (no- λ -correction). *Let $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ and let $R: \mathcal{S}^{\text{op}} \rightarrow \mathbf{Set}$ be any presheaf. A family $(x_a)_{a \in \text{Ob } \mathcal{S}}$ is a global section of R over $\mathcal{S}$ if and only if it is compatible with every $\mathcal{C}$ -morphism and every $\mathcal{D}$ -morphism. The matched-pair law λ plays no role.*

Proof. ($\Rightarrow$) Immediate: $\mathcal{C}$ - and $\mathcal{D}$ -morphisms are morphisms of $\mathcal{S}$.

($\Leftarrow$) Compatibility of a family with a morphism is closed under composition. If $R(g)(x_{\text{cod } g}) = x_{\text{dom } g}$ and $R(h)(x_{\text{cod } h}) = x_{\text{dom } h}$ with $\text{cod } h = \text{dom } g$, then by functoriality

$$R(g \circ h)(x_{\text{cod } g}) = R(h)(R(g)(x_{\text{cod } g})) = R(h)(x_{\text{dom } g}) = R(h)(x_{\text{cod } h}) = x_{\text{dom } h}.$$

Since every morphism of $\mathcal{S}$ is a word in $\mathcal{C}$ - and $\mathcal{D}$ -arrows — indeed $f = c \cdot d$ by the factorization — a family compatible with all $\mathcal{C}$ - and $\mathcal{D}$ -generators is compatible with all morphisms. The rewriting λ converts $\mathcal{DC}$ -words to $\mathcal{CD}$ -words but never changes the *morphism* it names, hence never changes the compatibility constraint. $\square$

Remark 3. The point is not that λ is trivial — it is generally nontrivial, and it is exactly what $[\omega]$ measures — but that it is inert *on sections*. The weld rearranges how a morphism is presented; a presheaf only sees the morphism. This is why no amount of weld structure can inject content into the inventory gluing problem, as long as $\mathcal{S}$ is a well-defined category at all.

Remark 4 (verification). Lemma 2 was brute-checked on a genuinely nontrivial Zappa–Szép product — the flip matched pair on two objects, $t_1 \cdot c = c \cdot t_0$ with $\lambda \neq \text{id}$ — over all functorial R valued in $\mathbb{Z}/3$: the global sections equalled the $\{\mathcal{C}, \mathcal{D}\}$ -generator-compatible families, in every case (`scratch/zs-presheaf_sections.py`).

Consequence. The set of consistent inventories is cut out entirely by the $\mathcal{C}$ - and $\mathcal{D}$ -constraints. Inventory consistency is exactly the degree-one amalgamation problem for the cover $\{\mathcal{C}, \mathcal{D}\}$ — which is what the next section computes.

4 Inventory consistency is a degree-one class

Theorem 5. *Let $\mathcal{S} = \mathcal{C} \bowtie \mathcal{D}$ (so the decomposition exists, $[\omega] = 0$) and let R be a transport local system of abelian stock-groups. Then global inventory is consistent if and only if $[\theta_R] = 0 \in H^1(\mathcal{S}; R)$. If moreover the factors are acyclic — $H^1(\mathcal{C}; R) = H^1(\mathcal{D}; R) = 0$, the “tree-like” procurement and delivery case — then Mayer–Vietoris for the cover $\{\mathcal{C}, \mathcal{D}\}$ gives the closed form*

$$H^1(\mathcal{S}; R) \cong \operatorname{coker}[H^0(\mathcal{C}; R) \oplus H^0(\mathcal{D}; R) \rightarrow H^0(\mathcal{O}; R)],$$

so the obstruction is exactly the reconvergence discrepancy: the failure of the node-potentials determined on each factor to agree on the shared nodes.

Proof. By Lemma 2 a global section is a family satisfying the $\mathcal{C}$ - and $\mathcal{D}$ -constraints. For a transport local system this says precisely that there is a node-potential φ with $\delta = d^0\varphi$, i.e. the 1-cochain δ is a coboundary, i.e. $[\theta_R] = [\delta] = 0$ in $H^1(\mathcal{S}; R)$. This equivalence uses only Lemma 2 and is unconditional on the factors.

For the closed form: the cover $\{\mathcal{C}, \mathcal{D}\}$ with discrete overlap $\mathcal{O}$ presents $\mathcal{S}$ as a pushout of categories glued along $\mathcal{O}$, then welded by λ , which by Lemma 2 is inert on cohomology of coefficient presheaves. The Mayer–Vietoris sequence of this cover reads

$$0 \rightarrow H^0(\mathcal{S}) \rightarrow H^0(\mathcal{C}) \oplus H^0(\mathcal{D}) \rightarrow H^0(\mathcal{O}) \rightarrow H^1(\mathcal{S}) \rightarrow H^1(\mathcal{C}) \oplus H^1(\mathcal{D}) \rightarrow \dots$$

(coefficients R throughout). The overlap $\mathcal{O}$ is discrete, so $H^{\geq 1}(\mathcal{O}; R) = 0$ and the two-piece cover with discrete overlap has no higher Čech terms; acyclic factors kill $H^1(\mathcal{C}) \oplus H^1(\mathcal{D})$. What remains is $H^1(\mathcal{S}; R) \cong \operatorname{coker}[H^0(\mathcal{C}) \oplus H^0(\mathcal{D}) \rightarrow H^0(\mathcal{O})]$. $\square$

Remark 6. Only the *closed form* needs acyclic factors. The equivalence “consistent $\iff [\theta_R] = 0$ ” holds for any factors, since it rests only on Lemma 2. For factors with their own loops, $H^1(\mathcal{S}; R)$ acquires the $H^1(\mathcal{C}) \oplus H^1(\mathcal{D})$ summand and the decomposition changes; the characterization does not.

Example 7 (the reconvergent diamond). Take the free category on the diamond quiver $a \rightrightarrows \{b, c\} \rightrightarrows d$: routes $a \rightarrow b \rightarrow d$ (arrows p, r) and $a \rightarrow c \rightarrow d$ (arrows q, s), kept distinct. The classifying space is the diamond graph, with $\beta_1 = 4 - 4 + 1 = 1$, so $H^1(\mathcal{S}; A) = A$ for any coefficient group A ; the single class is the loop-sum $\delta_p + \delta_r - \delta_q - \delta_s$. With transport deltas $(\delta_p, \delta_r, \delta_q, \delta_s) = (2, 1, 1, 3)$ the holonomy is $2 + 1 - 1 - 3 = -1 \neq 0$: the stock delivered at d differs by route, so inventory is inconsistent, and the class is the generator of H^1 . A coboundary $\delta = d^0\varphi$ is consistent (the diamond case is checked in `scratch/supply_chain_obstruction.py`). The failure is *degree one*, unambiguously. Over $\mathbb{Z}/n$ the loop-sum lives in $\mathbb{Z}/n$ and consistency is the vanishing of its residue; for a directed k -cycle with uniform step this is the familiar $H^1(C_k; \mathbb{Z}/n) = \mathbb{Z}/\gcd(k, n)$.

5 The two classes are independent

Theorem 8 (separation). *The classes $[\theta_R] \in H^1(\mathcal{S}; R)$ and $[\omega] \in H^2(\operatorname{Sk}_{\mathcal{S}}; \mathcal{D})$ are logically independent. Both of the following occur:*

- (W1) $[\omega] = 0$ with $[\theta_R] \neq 0$: the chain decomposes, yet inventory is inconsistent.
- (W2) $[\omega] \neq 0$ with $[\theta_R] = 0$: the chain does not decompose, yet inventory is trivially consistent.

Hence neither implication of (C) holds. Moreover the two classes live in different degrees and over different coefficient systems (R versus $\mathcal{D}$).

Proof. Witness W1 (breaks “ $\Leftarrow$ ”). Let $\mathcal{S}$ be the free category on the diamond quiver of Example 7. The quiver is directed-acyclic, so $\mathcal{S}$ has no non-identity invertible morphisms; the weld coefficient system is the trivial group, $\mathcal{D} = 0$, whence $H^2(\mathrm{Sk}_{\mathcal{S}}; \mathcal{D}) = H^2(\mathrm{Sk}_{\mathcal{S}}; 0) = 0$ and $[\omega] = 0$ is forced. (A free category is a strict factorization with nothing to obstruct.) Yet $\beta_1 = 1$, so $H^1(\mathcal{S}; R) = R$; choosing the transport system $(2, 1, 1, 3)$ of Example 7 gives $[\theta_R] \neq 0$ — inventory inconsistent.

Witness W2 (breaks “ $\Rightarrow$ ”). Let $\mathcal{S}$ be the rigid-twist category of [7]: a branch $a \rightarrow x \rightrightarrows y$ with $\mathrm{Hom}(y, y) = \mathbb{Z}/2$ and the twist $s_2 \circ p = (s \circ p) \cdot g$, for which $[\omega]$ is the generator of $H^2(\mathrm{Sk}_{\mathcal{S}}; \mathbb{Z}/2) = \mathbb{Z}/2$, so $[\omega] \neq 0$. Equip it with any coboundary inventory $\delta = d^0 \varphi$ — e.g. the zero transport system. Such an inventory is globally consistent by construction, $[\theta_R] = 0$, while $[\omega] \neq 0$.

Finally, in W1 the weld coefficient $\mathcal{D}$ is forced to 0 while the resource coefficient R is nonzero; the classes cannot even be compared by a natural map, let alone identified. Both witnesses are machine-checked in `scratch/separation_witnesses.py`. $\square$

Remark 9 (reading the two classes). $[\omega]$ answers “*does the supply chain even factor as procurement $\bowtie$ logistics, as the model asserts?*” — a structural precondition on the category. $[\theta_R]$ answers “*given the operations, does stock glue to a route-independent inventory?*” — the actual consistency question, about data on top of the category. Conjecture (C) conflated a precondition in degree two with the consistency class in degree one.

6 Where (C) came from, and why nothing rescues it

The conjecture is not arbitrary; it is a false generalization of blockchain re-entrancy, where $[\omega] = \varepsilon \in H^2 \cong \mathbb{Z}/2$ is the double-spend bit [8]. It is worth stating precisely why the template does not transfer.

Proposition 10 (no rescue regime). *There is no non-degenerate supply chain in which inventory consistency of a resource presheaf R literally equals $[\omega] = 0$. The appearance of equivalence in re-entrancy is not a special case of (C); it is a different problem, in which there is no separate resource presheaf at all.*

Proof. Inventory consistency is $[\theta_R] = 0$ in degree one with coefficients R (Theorem 5); $[\omega] = 0$ is a degree-two condition with coefficients $\mathcal{D}$. An equivalence would require a natural identification $H^1(\mathcal{S}; R) \cong H^2(\mathrm{Sk}_{\mathcal{S}}; \mathcal{D})$ of the two classes. No such identification exists in general: there is no suspension or dimension-shift relating a degree-one presheaf class to a degree-two weld class with different coefficients. For the free loop $\mathcal{S} = B\mathbb{Z}$ one has $H^1(B\mathbb{Z}; A) = A$ while $H^2(B\mathbb{Z}; A) = 0$ — the degrees simply do not match. And Theorem 8 already shows the two classes vary independently. $\square$

What re-entrancy actually is. In re-entrancy the conserved quantity *is the weld-state* — the composite protocol’s own state, living inside the matched pair, not on a separate presheaf of nodes. There $[\omega]$ is the whole story because there is nothing else: inventory equals weld-state by construction, so its consistency obstruction *is* the weld H^2 . Conjecture (C) imported this identification — “inventory = weld-state” — into supply chains, where it is false: supply-chain stock lives on the nodes, transported by operations, and is a genuinely separate degree-one datum.

Reconciliation with the $W_{n,\varepsilon}$ model

An earlier worked example [9] models “inventory inconsistency between two routes” by the warehouse family $W_{n,\varepsilon}$: objects Wh, Pr, De , with $s \cdot p = q$ and $s_2 \cdot p = q \cdot \tau^\varepsilon$, computing $[\omega] = \varepsilon \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathbb{Z}/n)$. This computation is *correct for the shape it uses* — but the shape decides which

of the two consistency notions it captures, and it is not the one the phrase “two routes” most naturally names.

Look at the geometry. In $W_{n,\varepsilon}$ the “two routes” are the *parallel* arrows s, s_2 sharing the same source and target (a rigid twist $a \rightarrow x \rightrightarrows y$); they differ by a vertex-group element τ^ε . A loop built from parallel arrows closes inside a single hom-set, so its class lives in the *vertex group*, hence in $H^2(\mathrm{Sk}; \mathcal{D})$. That is genuinely a weld phenomenon — whether the lot-cursor algebra closes into a distributive law. Call it *provenance consistency*; it is a legitimate degree-two question. Contrast the honest “two routes to a delivered good”: $a \rightarrow b \rightarrow d$ and $a \rightarrow c \rightarrow d$ through *distinct* intermediate nodes (the diamond, Example 7). Here the loop runs through the nerve of $\mathcal{S}$, not a single hom-set; its class lives in $H^1(\mathcal{S}; R)$. This is node-stock consistency — degree one.

So the dividing line is clean, decided entirely by *where the reconvergence loop lives*:

Reconvergence loop	Class	Reads
parallel arrows (vertex-group loop)	$H^2(\mathrm{Sk}; \mathcal{D})$	weld / provenance
distinct-node reconvergence (nerve loop)	$H^1(\mathcal{S}; R)$	node stock

The $W_{n,\varepsilon}$ computation therefore produced the weld-provenance class and labelled it “inventory inconsistency between two routes.” Under the natural reading of that phrase — distinct routes, reconvergent stock — the obstruction is H^1 , and Theorem 8 shows it is independent of the H^2 the example computed. The example is not wrong; it is a correct computation of the *other* class, and should be relabelled provenance consistency.

7 The degree classification

Conjecture (C) is false: inventory consistency is not the vanishing of the Zappa–Szép weld class. It is the vanishing of a degree-one class $[\theta_R] \in H^1(\mathcal{S}; R)$ on the resource presheaf — the route-reconvergence discrepancy — which is independent of the degree-two weld class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{S}}; \mathcal{D})$ in degree, in coefficients, and in truth value.

The honest result organizes compositional failure by cohomological degree, keyed to the locus of the conserved quantity.

Conserved quantity lives...	Failure class	Economic instance
on the nodes (transported by operations)	$[\theta_R] \in H^1$ — route discrepancy	supply-chain inventory
in the weld (the composite’s own state)	$[\omega] \in H^2$ — handoff order	blockchain re-entrancy
in the factorization existing at all	$[\omega] \in H^2$ — transversal closure	does the chain decompose?

Supply chains are therefore not a second instance of “failure = nonzero H^2 .” They are the first clean instance of “failure = nonzero H^1 ”: a route/reconvergence discrepancy, a distinct and arguably more common economic failure mode. The degree of the obstruction tells you where to look for the lost quantity.

Open boundary. Three honest limits.

1. *Torsor-valued inventory.* Lemma 2 and Theorem 5 are for inventory valued in a sheaf of *groups* (stock levels). If inventory is valued in *torsors* — stock defined only up to a gauge or allocation choice — gluing becomes a gerbe problem and the obstruction rises to H^2 .

A genuine degree-two inventory obstruction then exists; but it is a *higher-gauge* class with coefficients in R , still not equal to the weld $[\omega]$ (different coefficients). Characterizing it, and its relation to $[\omega]$, is the natural follow-up — very likely another “rhyme, not identity.”

2. *Non-acyclic factors.* The closed form in Theorem 5 assumes $H^1(\mathcal{C}) = H^1(\mathcal{D}) = 0$. For factors with their own loops the equivalence “consistent $\iff [\theta_R] = 0$ ” still holds (Lemma 2 is unconditional); only the explicit decomposition acquires the $H^1(\mathcal{C}) \oplus H^1(\mathcal{D})$ summand.
3. *The general comparison map.* Proposition 10 rules out a natural identification via explicit degree mismatch and the separation witnesses. The general question — when a connecting map in the descent spectral sequence of the $\{\mathcal{C}, \mathcal{D}\}$ cover relates node- H^1 to weld- H^2 — is sketched, not developed, and is not needed for the main theorems.

Fidelity. Throughout, “a supply chain is this category” is a modelling commitment, not a claim that the abstraction is uniquely faithful to any real logistics system; object-level fidelity remains open. The mathematics above concerns the abstraction we fixed in §2.

For the grant. The Applications/Impact story is sharpened, not weakened: two compositional failure modes, cleanly separated by cohomological degree, each with an economic witness and a machine-checkable obstruction. Correctness of composition has a price, and the degree of the obstruction tells you which account it is charged to.

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